

Data Transfer



The internet is the collection of data that is continuously shared with everyone. It is intriguing how the data travels from one place to another so quickly.



Whether it is for work or entertainment, we use the internet every single day. The only thing we require is a good internet connection. Devices need an internet connection to perform any task online.

Have you wondered how the internet works? All we see on our screens is a collection of words and images. They are made up of dots and pixels, which are made up of data. Since computers only understand code, everything needs to be converted before devices can send it. The code is then converted into electromagnetic waves. These travel from one place to another via internet cables. This whole process is called Data Transfer.

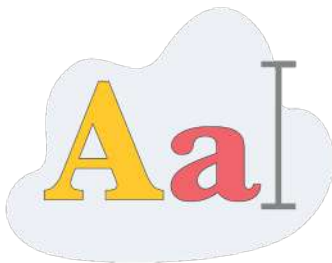
Computer network structure

1. Our personal computers are connected to a router.
2. The router is connected to the ISP (Internet Services Provider).
3. When we are online, we might connect to an app or website. The computer is then linked with the respective domain name server.
4. The server provides the information that we request and the data travels back to our computer.

The data transfer process comprises of 3 concepts:

1. Converting from pixel to code

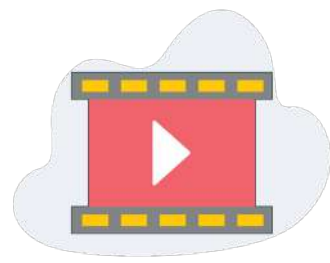
When we save an image on a computer, we can only see pixels. Even the words that we type are made of pixels. They emit light that is either on or off, depending on what is visible on the screen. The words we read are pixels translated from binary code.



Text



Image



Video

Data is stored in computers in binary language, which is a number system based on 0s and 1s. Each letter or number we see on a screen is represented by an 8-bit binary code.

Encoding data into binary and ASCII

ASCII Code: American Standard Code for Information Interchange is a widely used standard for encoding text documents on computers. Each letter is assigned a code from 0 to 127. They follow a decimal conversion simpler than binary.

Letter	ACSCII	Binary	Letter	ASCII	Binary
a	097	01100001	A	065	01000001
b	098	01100010	B	066	01000010
c	099	01100011	C	067	01000011
d	100	01100100	D	068	01000100
e	101	01100101	E	069	01000101
f	102	01100110	F	070	01000110
g	103	01100111	G	071	01000111
h	104	01101000	H	072	01001000
i	105	01101001	I	073	01001001
j	106	01101010	J	074	01001010
k	107	01101011	K	075	01001011
l	108	01101100	L	076	01001100
m	109	01101101	M	077	01001101
n	110	01101110	N	078	01001110
o	111	01101111	O	079	01001111
p	112	01110000	P	080	01100001
q	113	01110001	Q	081	01100001
r	114	01110010	R	082	01100001
s	115	01110011	S	083	01100001
t	116	01110100	T	084	01100001
u	117	01110101	U	085	01100001
v	118	01110110	V	086	01100001
w	119	01110111	W	087	01100001
x	120	01111000	X	088	01100001
y	121	01111001	Y	089	01100001
z	122	01111010	Z	090	01100001

a. Converting decimal to binary

A number consists of two aspects – Value and Base. The value of a number is the digit that it represents. A digit will have two values: face value and place value.

- A face value is the number itself.

For example: In 15, the face value of 1 is 1, and 5 is 5.

- A place value is defined by the order or power of the base a number is on. Since we use 10 digits (0-9) the base is 10.

For example: In 15, the place value of 1 is 1×10^1 and 5 is 5×10^0 .

Base

Base is the number of digits that are used to represent a number. We use base 10 since we use the digits 0, 1, 2, 3, 4, 5, 6, 7, 8, 9. These are 10 Hindu-Arabic digits of a number system.

Similarly, there can be base 2, 3, 4, and so on.

- Base 2 system is called Binary number system with digits 0-1.
- Base 8 system is called Octal number system with digits 0-7.
- Base 16 system is called Hexa-decimal number system, with digits 0-9 and A-F.

Each system has different applications, which we will study later.

Let's take a number like 156 at the base 10.

We want to convert it to base 2. We will keep dividing 156 by 2 and record the remainders.

Once we have all the remainders, we write them in reverse order. The number that we get is the converted binary code.

In this example, $156_{10} = 10011100_2$

b. Converting binary to decimal

To convert from binary to decimal, we consider eight powers of 2.

2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0
128	64	32	16	8	4	2	1

Any number can be broken down into a sum of the powers of 2.

For example: $179 = 128 + 32 + 16 + 2 + 1$

1	0	1	1	0	0	1	1
128	64	32	16	8	4	2	1

Whichever power of 2 is present in the number breakdown, we mark it as 1. The powers that are missing are marked as 0. Then we write them from left to right.

$179_{10} = 10110011_2$

Activity 1

Find the binary values for the given ASCII codes.

Number	0	1	2	3	4
ASCII	48	49	50	51	52
Binary	00110000		00110010	00110011	00110100
Number	5	6	7	8	9
ASCII	53	54	55	56	57
Binary	00110101		00110111	00111000	

Exercise

Convert from decimal to binary

- 1 125 _____
- 2 400 _____
- 3 88 _____
- 4 16 _____

Convert from binary to decimal

- 5 01101110 _____
- 6 11011011 _____
- 7 11111111 _____
- 8 10001000 _____



Recall:

Conversion scale:

1 Byte (B)	8 Bits
1 Kilo Bytes (KB)	1024 Bytes
1 Mega Bytes (MB)	1024 Kilo Bytes
1 Giga Bytes (GB)	1024 Mega Bytes
1 Tera Bytes (TB)	1024 Giga Bytes

Activity 3

Use the internet to find the ASCII and binary codes for the following:

Letter	space	!	“	#	\$	%	&	‘
Binary								
ASCII								
Letter	(	)	*	+	,	-	.	/
Binary								
ASCII								

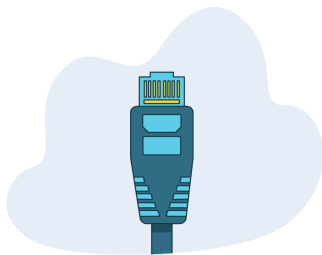
Activity 3

Write your name in ASCII and binary code. Use the correct code for each letter.

2. Internet connection cables

A computing device can be connected to the internet via one of these three connectors:

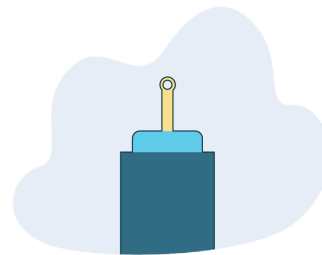
DIFFERENT TYPES OF INTERNET CONNECTION CABLES



ETHERNET



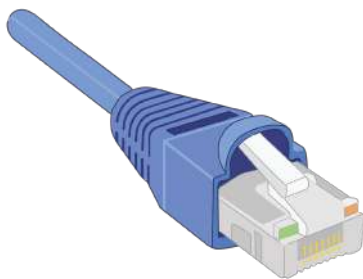
DSL



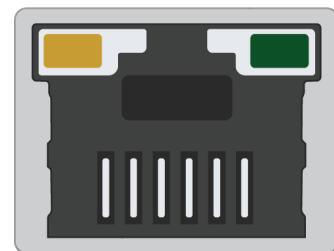
FIBER OPTIC

a. Ethernet cable (RJ-45)

An ethernet cable connects a computer to a router. This cable is connected via a computer's Ethernet port and allows the user to access the internet.



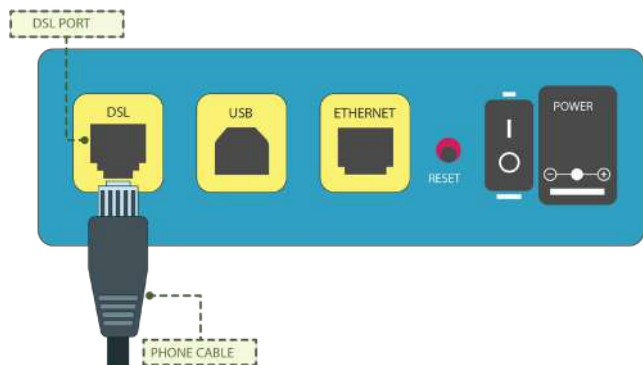
Ethernet Cable



Ethernet Port

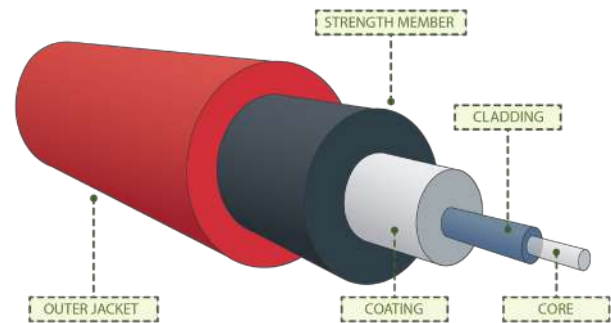
b. DSL cable (RJ-11)

DSL stands for Digital Subscriber Line. It is a type of connector that transmits data over a telephone network through a telephone cable. The DSL internet depends on the user's distance from the ISP.



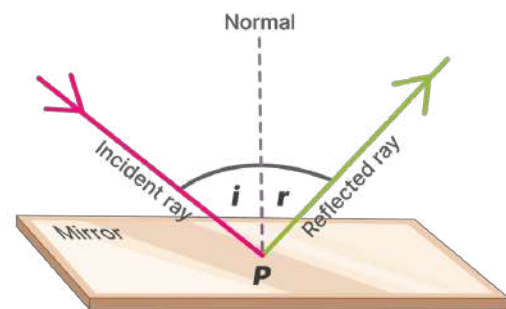
c. Fiber optic cable

A fiber optic cable is made of thin, glass fibers. They carry data in the form of light. Light is transmitted down the fibers in LED or laser pulses that travel extremely fast. These pulses carry binary data that makes up everything we see on the internet.

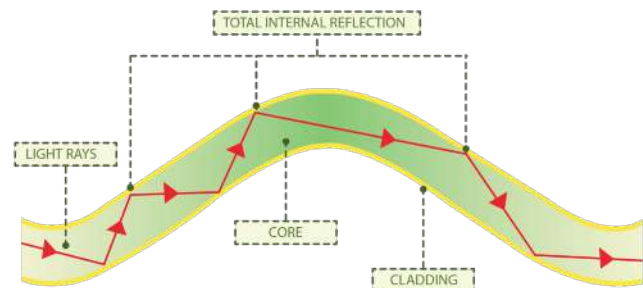


What happens inside an optical fiber?

Light falls on and reflects off a mirror at an equal angle. The light bounces back at an equal angle of reflection. This is the nature of light; it always travels in a straight line.



In the case of an optical fiber, a unique phenomenon occurs. Light reflects and bends inside the tube without escaping it. Even after it reflects repeatedly, most of the light is contained inside the tube. This phenomenon is called Total Internal Reflection. The light carries data encoded in binary, transferring it from the source to its destination.



Comparison between DSL, ethernet, and fiber optic cables

	Cost	Actual Download Speeds	Actual Upload Speeds	Range of Connection Speeds
DSL	Low	5-6 Mbps	0.4 – 1.2 Mbps	None
Ethernet	Medium Low	2-50 Mbps	2-10 Mbps	Medium
Fiber Optic	Medium	5-10,000 Mbps	5-10,000 Mbps	High

The three modes of connectivity show the evolution of internet connectivity hardware.

DSL → Ethernet → Fiber Optic Cable

- A fiber optic cable is the most expensive but has the highest data transfer speed.
- DSL is the cheapest and has the lowest speed.
- A DSL cable is plugged into a telephone line, making it impossible to use a landline phone and the internet on the same line.

3. Packets as data carriers

The computers we use are a part of a Local Area Network (LAN). Many such LANs come together and form a Wide Area Network (WAN). Many WANs come together to make up the internet.



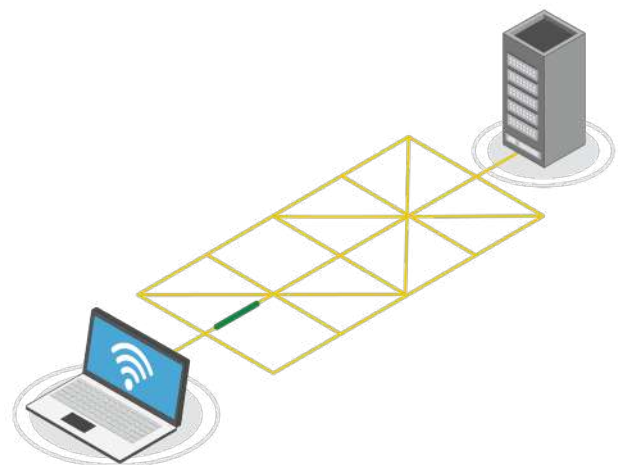
PAN: Personal Area Network, operates in a house

LAN: Local Area Network, operates on a society or neighbourhood scale

MAN: Metropolitan Area Network, operates on a city or town scale

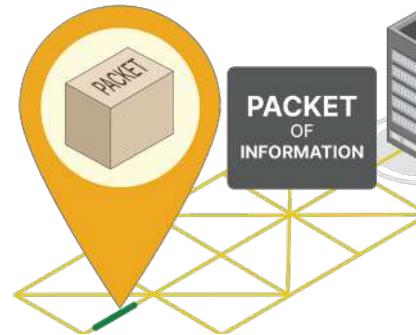
WAN: Wide Area Network, operates on a country or continent scale

Billions of devices are connected to each other over the internet. Data sent to and from devices uses packets to travel and reach the correct destination. Data is stored inside a packet and sent to the receiver following a route. If the primary route is busy, packets take an alternative route.



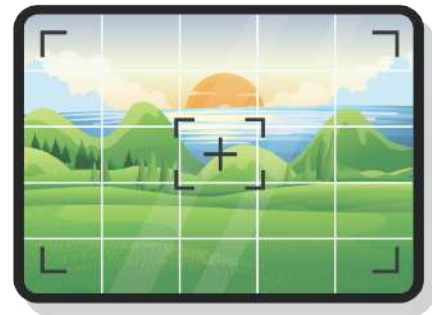
Often, the size of data is huge. In this case, it needs to be broken down into batches. Packets carry these batches, and each batch can follow any route to reach the destination. After all the batches reach the destination, they are assembled back together.

Every packet contains information about the sender and receiver's IP address.



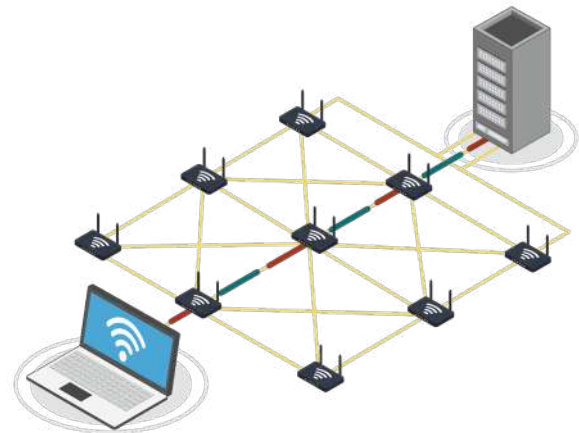
For example:

When we send a large file, like an image, it is broken down into small chunks of data. Each chunk has its own code. Individual chunks are sent in the form of packets.



Routers forward data packets between networks. Every time a packet comes in, there is a record of which routers it has visited. Every router keeps track of information passing through each path. It chooses the path with the lowest congestion to forward the packet.

When packets have multiple path options for travel, it makes a stable network.



When this data reaches the device that requested it, it is reassembled. It gets restored and makes a complete image before it is displayed to the user.

Text, videos, and sound follow the same method as well. The speed of this whole process is so fast that we do not know when all of this is happening.

Exercise

Fill in the blanks

- 9 Fiber optic cables work on the principle of _____.
- 10 The internet network that operates on a society or neighbourhood scale is _____.
- 11 _____ is widely used standard to encode text documents on computers.
- 12 Packets carry data as well as the _____ of the sender and the receiver.

The whole process of data transfer starts with the conversion of pixels into data. This data is sent to the receiver via a cable. This cable can be a Fiber cable, Ethernet cable, or a DSL cable. The data is broken into several packets based on its size. Once they reach the receiver, they are reassembled.

Millions of data packets ensure quick data transfer that helps us access the information anytime from anywhere.



Tech Flash

- **ISP** : An Internet Service Provider is a company or organisation that provides services to end-users to connect to the internet. To get a new connection, we contact an ISP provider.
- **Number System** : The system of naming or representing numbers. Every number system has a base, which refers to the number of digits used to represent numbers. We commonly use base 10 which means the digits used are 0-9.
- **Binary** : Number system with base 2, digits used are 0,1.
- **Octal** : Number system with base 8, digits range from 0-7.
- **Hexadecimal** : Number system with base 16 and digits ranging from 0-9 and A-F.
- **Packet** : A packet is a data carrier. Along with data, it carries the address, and other information which it requires to reach its destination.
- **ASCII (pronounced - askee)** : The American Standard Code for Information Interchange is a character encoding standard for electronic communication.